

Smit Malde

[linkedin.com/in/smit-malde](https://www.linkedin.com/in/smit-malde) | smit334@berkeley.edu | smitmalde.xyz

EDUCATION

University of California, Berkeley

May 2025

B.S. in Electrical Engineering and Computer Sciences

- **Technical Skills:** PCB design, C, Python, Java, Solidity, Git, NumPy, SQL, Photoshop, Altium, Adobe XD.
- **Technical Coursework:** Power Electronics, Microelectronic Devices, Computer Security, Computer Architecture and Machine Structures, Designing Information Devices and Systems, Signals and Systems, Data Structures, Structure and Interpretation of Computer Programs, Techniques of Data Science, Discrete Mathematics and Probability, Berkeley Method of Entrepreneurship.
- **Projects:** Made a Voice-controlled Car using Arduino and a voice-word Classifier Model. Built a Git inspired Version Control System, Programmed an Encryption software based on the Enigma Machine, ML classifier to identify movie genres, created a game with an AI opponent using game tree algorithms, Scaled a Niche Wildlife Instagram account to 200,000+ followers.
 - Designed a PCB to Bring-Up and boot Linux on a custom Tapped out 16nm Chip.

Oshwal Academy Nairobi

Class of 2021

Further Pure Mathematics, Physics, ICT, Chemistry

AAAA

EXPERIENCE

Tesla

Palo Alto, California

Electrical Design Engineer – Battery Electronics

August 2025 – Present

- Design and validate Battery Management System (BMS) boards and high voltage controllers, from schematic capture and layout through bring-up and validation.
- Apply AI across the hardware workflow, including AI-driven test automation that accelerates bench validation.

Formula Electric at Berkeley

Berkeley, California

Chief of Electrical Engineering and Computer Science

September 2022 – June 2025

- Led a team of over 50 Engineers for the development of all Low Voltage and High Voltage Electrical Systems, Architecture and Firmware/Software for the Formula style Electric race car to compete in FSAE.
- Designed a PCB for a High voltage (600V) to low voltage (12V) DC-DC galvanically isolated converter using the Flyback topology to power the low voltage system of the car through the accumulator.
- Designed a PCB for the battery management system for the accumulator which uses passive cell balancing while charging.

Tesla

Palo Alto, California

Electrical and Firmware Engineer Intern – Battery Electronics

January 2024 – August 2024

- Designed and brought up a 24 Channel BMS PCB interfaced through isoSPI.
- Designed and brought up a Thermistor Characterization and Reliability Testing board utilizing BMS ASICs.
- Designed solutions for reducing noise emitted by high frequency communication, with an added benefit of reducing manufacturing cost per vehicle.

Rivian

Irvine, California

Electrical Engineer – Electronics Hardware

May 2023 – August 2023

- Designed Cell Simulator boards with 1.22mV precision, capable of stacking to create a modular arrangement of up to 18 cells in series to test various aspects of Electrical Control Units in the R1 series of cars.
- Designed and collaborated on a Test Automation framework containing Cell Simulator, Analog and Digital I/O and Constant Current customizable E-load cards to test various PCBs and worked on testing for next Gen architecture updates.

Make A Move Charity Organization (Non-Profit Organization)

Nairobi, Kenya

Co-Founder and Treasurer

July 2017 – Present

- Pitched the situation of rural children to sponsors and non-profits in Kenya to raise over 2000 USD for their support.
- Aided students in rural areas and reduced the high school dropout rate for schools by holding educational workshops.

ACTIVITIES

Duke of Edinburgh Award (President's Award)

Nairobi, Kenya

Club Chairperson and Gold Award Holder

September 2015 – May 2020

- Received the Bronze, Silver, and Gold awards by successfully completing rigorous hiking expeditions throughout Kenya.
- As the Chairperson of the club, I led multiple groups of 100+ participants to complete Bronze and Silver expeditions.

Kenya National Swimming Team

Nairobi, Kenya

Team Member

2013 – 2019

- Represented and won 3 medals for Kenya in FEASSSA.
- National record holder in 12-13 boys and 14-15 boys 50m Backstroke and 12-13 boys 100m Backstroke.
- Selected to represent Kenya in the Commonwealth Games.

AWARDS & LANGUAGES

Awards: The Duke of Edinburgh Award (Bronze, Silver & Gold), O-level Honors, A-level Honors

Languages: English (Native Proficiency), Gujarati (Native Proficiency), Hindi (Fluent) and Swahili (Fluent).